

**GFS SPECIFICATION**

**DESIGN AND CONSTRUCTION OF CONCRETE  
STRUCTURES FOR ONSHORE LNG FACILITIES**

GFS IG.34.85.31-Gen.

February 2018

ECCN EAR99

**GENERAL FUNCTIONAL STANDARD**



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## 1. INTRODUCTION

### 1.1 SCOPE

This new GFS specifies requirements and gives recommendations for the design and execution of onshore concrete structures for projects in Integrated Gas (IG) facilities. This GFS applies to both in-situ work and construction using manufactured/ prefabricated concrete elements. It specifies the industry codes, standards and additional technical requirements (including safety-related requirements) that comprise the minimum technical requirements to be applied to the IG facilities that are covered by the scope of this GFS.

This GFS is classified as a TIER 2 document. See (1.3) for definitions of TIER 1/TIER 2.

### 1.2 DISTRIBUTION, INTENDED USE AND REGULATORY CONSIDERATIONS

Unless otherwise authorised by Shell GSI, the distribution of this GFS is confined to Shell companies and, where necessary, to Contractors nominated by them. Any authorised access to a GFS does not for that reason constitute an authorisation to any documents, data or information to which the GFS may refer.

This GFS is intended for use in LNG production facilities and associated supply/marketing. This GFS may also be applied in other similar facilities.

When GFSs are applied, a Management of Change (MOC) process shall be implemented; this is of particular importance when existing facilities are to be modified.

If national and/or local regulations exist in which some of the requirements could be more stringent than in this GFS, the Contractor shall determine by careful scrutiny which of the requirements are the more stringent and, which combination of requirements will be acceptable with regards to safety, environmental, economic and legal aspects. In all cases, the Contractor shall inform the Principal of any deviation from the requirements of GFS which is considered to be necessary in order to comply with national and/or local regulations. The Principal may then negotiate with the Authorities concerned the objective being to obtain agreement to follow this GFS as closely as possible.

### 1.3 DEFINITIONS

#### 1.3.1 General definitions

The **Contractor** is the party that carries out all or part of the design, engineering, procurement, construction, commissioning or management of a project or operation of a facility. The Principal may undertake all or part of the duties of the Contractor.

The **Manufacturer/Supplier** is the party that manufactures or supplies equipment and services to perform the duties specified by the Contractor.

The **Principal** is the party that initiates the project and ultimately pays for it. The Principal may also include an agent or consultant authorised to act for, and on behalf of, the Principal.

**TIER 1** packaged equipment solutions are kept evergreen and have a revision cycle.

**TIER 2** packaged equipment solutions do not have a revision cycle and updated when needed.

The word **shall** indicates a requirement.

The word **should** indicates a recommendation.

The word **may** indicates a permitted option.

**1.3.2 Abbreviations**

| <b>Term</b> | <b>Definition</b>                         |
|-------------|-------------------------------------------|
| <b>ITP</b>  | Inspection and Test Plan                  |
| <b>IG</b>   | Integrated Gas                            |
| <b>LNG</b>  | Liquefied natural gas                     |
| <b>PTI</b>  | Post-Tensioning Institute                 |
| <b>QCR</b>  | Quality Control Records                   |
| <b>SI</b>   | International System of Measurement Units |
| <b>USC</b>  | US Customary Measurement Units            |

**1.4 CROSS-REFERENCES**

Where cross-references to other parts of this GFS are made, the referenced section or clause number is shown in brackets ( ). Other documents referenced by this GFS are listed in (7).

**1.5 SUMMARY OF MAIN CHANGES**

This is a new GFS.

**1.6 COMMENTS ON THIS GFS**

Comments on this GFS may be submitted to the Administrator using the same methods as for submitting comments on DEPs:

|                                                                      |                                                                                                                                                                                                                                                               |
|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Shell DEPs Online</b><br>(Users with access to Shell DEPs Online) | Enter the Shell DEPs Online system at <a href="https://www.shelldeps.com">https://www.shelldeps.com</a><br>Select a GFS and then go to the details screen for that GFS.<br>Click on the "Give feedback" link, fill in the online form and submit.             |
| <b>DEP Feedback System</b><br>(Users with access to Shell Wide Web)  | Enter comments directly in the DEP Feedback System which is accessible from the Technical Standards Portal <a href="http://www.shell.com/standards">http://www.shell.com/standards</a> .<br>Select "Submit DEP Feedback", fill in the online form and submit. |
| <b>DEP Standard Form</b><br>(other users)                            | Use Standard Form PES 00.00.05.82-Gen. to record feedback and email the form to the Administrator at <a href="mailto:standards@shell.com">standards@shell.com</a> .                                                                                           |

Feedback that has been registered in the Feedback System by using one of the above options will be reviewed by the GFS Custodian for potential improvements to the GFS.

**1.7 DUAL UNITS**

This GFS contains both the International System (SI) units, as well as the corresponding US Customary (USC) units, which are given following the SI units in brackets. When agreed by the Principal, the indicated USC values/units may be used.

#### 1.8 NON NORMATIVE TEXT (COMMENTARY)

Text shown in italic style in this GFS indicates text that is non-normative and is provided as explanation or background information only.

*Non-normative text is normally indented slightly to the right of the relevant GFS clause.*

#### 1.9 UPDATE POLICY

This GFS will be updated when applied in a project in accordance with the framework for TIER 2 Packaged Equipment Solutions.

## **2. GENERAL**

### **2.1 PURPOSE OF THIS GFS**

1. This GFS provides the basis for the development of Project Specifications and supporting documentation for design, material selection and construction of onshore concrete structures for projects based on the National Standards supplemented by the requirements of this GFS.

### **2.2 APPLICABLE CODES AND STANDARDS**

1. The sections in this GFS are based on American Standards and are the minimum requirements for design and construction of onshore concrete structures.

## **3. DESIGN OF CONCRETE STRUCTURES**

### **3.1 DESIGN REQUIREMENTS OF CONCRETE STRUCTURES**

1. The design and calculations of concrete structures shall be in accordance with ACI 318.
2. Loads and load combinations for all onshore concrete structures shall be in accordance with GFS IG.34.85.02-Gen.

### **3.2 CONSTRUCTION AND EXPANSION JOINTS**

1. Construction and expansion joints shall be clearly specified and indicated on the concrete design drawings.
2. Construction joints shall be perpendicular to the main reinforcement and located away from points of peak demand so as to least impair the performance of the structure.
  - a. Provision shall be made for the transfer of shear and other forces through the construction joint.

### **3.3 ANCHOR BOLTS**

1. Anchor bolts design shall be in accordance with GFS IG.34.85.20-Gen.

### **3.4 LIQUID RETAINING STRUCTURES**

1. Liquid retaining structures shall be designed in accordance with ACI 350/350R.
2. Crack width limits shall be  $\leq 0.1$  mm (0.004 in) in accordance with ACI 224R.
3. Crack width calculations shall be determined in accordance with the limit state principles and formulae presented within Eurocode 2.

### **3.5 PRE-STRESSED CONCRETE STRUCTURES**

1. Pre-stressed concrete shall only be allowed for piles, bridges/culvert decks in roads and concrete LNG tanks.
2. Design of pre-stressed concrete shall be in accordance with ACI 318 and PCI TR-7.
3. Anchorage device and local zone reinforcement shall meet the requirements of AASHTO Division I, Articles 9.21.7.2.2 through 9.21.7.2.4.

### **3.6 PRECAST CONCRETE**

1. Precast concrete members and connections shall be designed in accordance with ACI 318, ACI 550.2R and PCI TR-7.

#### **4. MATERIALS AND COMPONENTS**

##### **4.1 ADMIXTURES**

1. Concrete admixtures shall be in accordance with ASTM C494/C494M.
2. Only one Manufacturer/Supplier of each admixture shall be used throughout the course of the project.
3. Calcium chloride and admixtures containing soluble chlorides shall not be permitted.

##### **4.2 AGGREGATES**

1. Aggregates shall:
  - a. be obtained from a single source throughout the course of the project;
  - b. be in accordance with ASTM C33;
  - c. not be "gap-graded";
  - d. be free from dust, loam, dirt, vegetable matter or other foreign substances deleterious to concrete.

##### **4.3 ANCHOR BOLTS AND EMBEDMENTS**

1. Anchor bolt materials shall be in accordance with GFS IG.34.85.20-Gen.
2. Embedded steel material shall be as per ASTM A36/A36M and hot-dip galvanised (except the anchorage designed based on embedment length) in accordance with §5.3 of GFS IG.34.85.21-Gen.
3. Aluminium shall not be used for embedment materials.

##### **4.4 CEMENT**

1. Only one brand of cement shall be used throughout the course of the project.
2. For Hydrocarbon scope (e.g., Process units or storage) and Mass concreting (5.9) cement shall be:
  - a. Blended Portland Blast Furnace Cement with minimum 65 % blast furnace slag complying with the requirements of ASTM C595/C595M.
    - i. Slag shall be compliant with ASTM C989.
  - b. Subject to approval by the Principal, blended cement with fly ash shall be used, provided the Contractor demonstrates that the obtained concrete mix has same durability, impermeability and resistance to prevailing environmental conditions as the concrete mix based on Blended Portland Blast Furnace Cement with minimum 65% blast furnace slag.
3. Fly ash shall be compliant with ASTM C618 Type C or Type F.
  - a. If ASTM C618 Type C fly ash is used in a moderately to extremely aggressive sulphate environment (as defined in Table 4.3.1 of ACI 318-05), the mix design shall be tested in accordance with ASTM C 1012 to verify sulphate resistance.

##### **4.5 JOINT FILLER**

1. Remoulded expansion joint filler shall conform to ASTM D994 or ASTM D1751.



#### 4.6 JOINT SEALANT

1. Sealer shall be chemical resistant (regarding possible spills) two-component polysulfide compound.
2. Sealer for down hand application shall be liquid self-levelling grade and in accordance with ASTM D6690.
3. Sealer for vertical joints or for overhead application shall be trowel grade and in accordance with ASTM C881/C881M Type IV Grade Grade-3, Class-B/C.

#### 4.7 MIXING WATER

1. The water used for mixing water, ice, curing, or any other function relating to the placement of concrete shall be in accordance with ACI 318 and ASTM C94 and not contain matter which could be harmful to the reinforcement or the durability of the concrete and to the hydration of cementitious material in the concrete mix.

#### 4.8 REINFORCING STEEL AND PRE-STRESSING TENDONS

1. Reinforcement for concrete work shall be in accordance with ACI 318.
2. Reinforcement shall be hot rolled; high yield deformed bars.
3. Reinforcing bars shall be uncoated deformed billet steel.
4. Welded wire fabric shall be flat (unrolled) plain wire fabric.
5. Tie wire shall be black annealed wire, of 1.5 mm diameter (16 BWG).
6. Pre-stressing tendons shall be the type and strength required by the project specifications and conform to one of the following:
  - a. ASTM A 416M;
  - b. ASTM A 421M;
  - c. ASTM A 722M;
  - d. ASTM A 779M;
  - e. ASTM A 882M.
7. In areas where reinforcing steel congestion is anticipated, mechanical couplers can be substituted to enable full development of reinforcing strength at splices.
  - a. Use of mechanical couplers shall be subject to the approval of the Principal.

#### 4.9 GROUT

1. Material for grouting of structural and equipment base shall be as per GFS IG.34.85.20-Gen.

**4.10 VAPOUR BARRIER/ BUILDING PAPER**

1. Vapour barrier/building paper shall be a virgin heavy-duty polyethylene film and have the following characteristics:
  - a. a minimum thickness of 0.15 mm (0.006 in);
  - b. uniform thickness throughout;
  - c. overlap not less than 150 mm (6 in) at all joints and intersections;
  - d. flexible to enable construction at temperatures down to the lowest possible site temperatures;
  - e. moisture transmission of less than 0.45 metric perms (0.3 US perms), determined in accordance with ASTM E96/E96M;
  - f. free of any pinholes, nozzle scars or other imperfections.

**5. EXECUTION OF CONCRETE STRUCTURES****5.1 GENERAL**

1. Concrete construction shall be in accordance with ACI 301 and ACI 318.
2. Tolerances for concrete surfaces shall be in accordance with ACI 117.

**5.2 EXECUTION DOCUMENTS**

1. The Contractor shall propose and submit the general notes and Project Specification for Concrete works for the review and approval by the Principal.
2. Work shall not commence until the concrete design drawings have been approved by the Principal.

## 5.3 CONCRETE MIX DESIGN

1. The concrete mix design shall be submitted to the Principal for review and approval.
2. Self-compacting concrete shall not be used.
3. Fibre reinforced concrete shall only be used for non-structural applications, such as paving, in accordance with ACI 544.1R.
4. The compressive strength of the concrete mix shall meet both of the following:
  - a. be in accordance with the concrete design drawings and Project Specifications;
  - b. have a minimum strength after 28 days as the applicable item in Table 5.1.

**Table 5.1 Concrete mixing and strengths**

| Item                                                                                                                                                           | Design compressive strength of concrete |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|                                                                                                                                                                | Per ACI 318                             |
| Slabs, area paving, roads, cover slabs for cable trenches, watertight concrete, machinery foundations, and concrete exposed to freezing and thawing conditions | 28 MPa (4000 psi)                       |
| Structures and associated foundations, stationary equipment foundations with the exception of jetties                                                          | 28 MPa (4000 psi)                       |
| Seal slab or blinding concrete, and miscellaneous non-structural concrete fill.                                                                                | 14 MPa (2000 psi)                       |
| Concrete envelopes for underground electrical ducts                                                                                                            | 14 MPa (2000 psi)                       |
| Pre-stressed concrete structures and reinforced concrete of marine structures                                                                                  | 41 MPa (6000 psi)                       |

5. The required concrete mixes shall be designed in accordance with ACI 301 and ASTM C94/C94M for Ready-Mix Concrete.
6. Performance and acceptance criteria of concrete mixes shall be as per ACI 318.

7. The maximum Water-Cement Ratio (WCR) shall be as per the applicable ratio in Table 5.2.

**Table 5.2 Maximum Water-Cement Ratio**

| Structure                                                                                       | WCR maximum |
|-------------------------------------------------------------------------------------------------|-------------|
| Typical Structures (except as Note 1)                                                           | 0.45        |
| Seal Slab or Blinding concrete and miscellaneous non-structural concrete fill.                  | 0.55        |
| liquid-retaining (hydraulic) structures and marine structures                                   | 0.40        |
| Concrete deposited by tremie underwater or for pre-stressed concrete structures                 | 0.44        |
| Note 1: For mixtures containing pozzolan, the WCR is the water to (cement plus pozzolan) ratio. |             |

#### 5.4 FALSEWORK AND FORMWORK

1. All formwork, falsework and re-shoring (formwork) shall be in accordance with ACI 301, ACI 304R and ACI 347R.
2. Chamfer strips of 20 mm (3/4 in) shall be provided at all corners on permanently exposed surfaces.
3. A formwork striking plan shall be produced by the Contractor, which considers the climatic conditions, the heat of hydration, strength development, load conditions and relevant curing methods.
4. Contractor shall develop design calculations for formwork supporting the concrete weight.

#### 5.5 REINFORCEMENT FABRICATION

1. Welding of reinforced steel shall not be permitted.
2. Reinforcement steel shall be bent in the cold condition.
  - a. Reinforcement bars which have been bent and then straightened shall not be incorporated in the main works.
3. Reinforcement drawings and bending schedules shall be prepared in accordance with ACI SP-66.
4. Reinforcing bars shall be stored to avoid contamination with oil/ grease and raised above the ground (on wooden sleepers) to avoid contact with soil.

**5.6 PLACEMENT****5.6.1 Placing reinforcement**

1. Exposed reinforcing bars including dowels that are vertical shall be protected using plastic protection caps.
2. All reinforcing shall be in place prior to pouring concrete.
  - a. No wet placing of reinforcement (i.e. during concreting operations) shall be permitted.
3. Minimum concrete cover shall be in accordance with ACI 318.
  - a. For concrete exposed to hydrocarbons and prone to chemical attacks minimum cover shall be 50 mm (2 in).
4. Concrete spacer blocks with identical compressive strength with concrete or hard plastic blocks shall be used.
5. At the time of concrete placement, reinforcing steel shall be free from hardened mortar, loose flaky rust, mud, oil, or other coating which may be detrimental to bonding with the concrete.

**5.6.2 Placement of embedments**

1. Anchors, inserts, anchor rails, sleeves, drains, curb and seat angles, nosing, and other embedded items shall be positioned or templated in and securely anchored before placing concrete.
  - a. Welding of these items to the reinforcing bars shall not be permitted.

**5.6.3 Concrete placement**

1. The concrete temperature at the point of discharge shall not exceed 32 °C (90 °F).
2. Mixing of concrete shall not be done in concrete trucks.
3. Water shall not be added to increase slump.
4. Concrete shall not be hand-mixed or re-tempered.
5. No additives or ingredients shall be added to the concrete after batching.
6. Any proposed change in mix design during the concrete work shall be subject to approval by the Principal.
7. Concrete that has achieved initial set or has been contaminated by foreign matter shall not be deposited in the structure or foundation.
8. Concrete shall not be placed during rain, sleet, or snow unless protection is provided to avoid any changes in concrete design and material properties.
9. For underwater structures, prior to the pour, a concrete placing plan shall be submitted to the Principal for review and approval.

*Concrete trials to demonstrate the required quality and consistency of the concrete mix are required for underwater concrete pours.*
10. For placing by pumping, the concrete mix shall be in accordance with ACI 304.2R.
11. A contingency plan (including facilities), to ensure the continuity of concrete placing in case the pump fails, shall be submitted for the Principal review and approval.
12. All temperature/slump measurement and samples for compressive strength test shall be taken from the end of the discharge line.
  - a. All samples for compressive strength test shall be taken between 15 % and 85 % delivery of a truck.

13. All type of grout shall be constructed and placed in accordance with GFS IG.34.85.20-Gen.

#### **5.6.4 Placement of pre-fabricated concrete**

1. When prefabricated concrete elements are placed directly on steel structures, a load bearing layer shall be placed between those elements.
  - a. The load-bearing layer material shall be neoprene, nylon, rubber or equivalent as specified by the Principal.
  - b. The load-bearing material shall be chemical resistant when exposed to hydrocarbons.
2. When prefabricated elements are placed in in-situ concrete slots, non-shrink grout shall be used to provide a bearing surface in the slot in accordance with the grout Manufacturer/Supplier's specifications.

#### **5.6.5 Placement of water-stops**

1. Water-stops shall be installed and located according to the Manufacturer/Supplier's recommendations, specifications and drawings as part of the formwork installation.
2. Water stops shall be oil resistant; non-metallic materials such as neoprene, styrene butadiene rubber, or similar, used for sealing joints in concrete.
3. Water-stop joints shall develop effective water tightness equal to that of the continuous water-stop material.
4. Water-stop material shall be adequately supported during concrete placement.

#### **5.7 JOINTS**

1. All joints shall be located and constructed in accordance with the concrete design drawings and Project Specifications.
2. Control joints shall be installed in accordance with ACI 302.1R.

#### **5.8 TRIAL MIXES AND TRIAL POURS**

1. Trial mixes in batching plant and trial pours shall be conducted prior placing of permanent works to demonstrate the following:
  - a. required durability/exposure condition;
  - b. desired workability of the design mix concrete;
  - c. that the specified concrete grade can be achieved.
2. For each trial mix, inspection and testing shall be performed in accordance with the ITP.
3. The time required for any trial mixes and pours with associated testing shall be included as a part of the construction schedule, being completed one month prior to the commencement of concreting works.
4. Method statements for both trial mixes and trial pours shall be developed and submitted to the Principal for review and approval.

## 5.9 MASS CONCRETING

1. The requirements of Section (5.9) shall be applied to concrete sections that meet the following criteria:
  - a. For concrete pours subject to limited external restraining, such as foundations cast into blinding and pile caps with thickness more than 1200 mm (4 ft);
  - b. Where concrete members are subject to high external restraining, such as concrete walls on hardened slabs, sections with thickness more than 500 mm (20 in).
2. Mass concreting shall be in accordance with ACI 207.1R, ACI 207.2R and ACI 207.4R.
3. The temperature of concrete at the point of placement shall be in accordance with Section 8.3 of ACI 301.
4. The maximum peak temperature in the inside of the concrete pour shall not exceed 70 °C (160 °F).
5. The temperature gradient between the inside and outside faces of the concrete shall not be more than:
  - a. 20 °C (36 °F) for concrete pours subject to limited external restraining;
  - b. 15 °C (27 °F) for concrete members subject to high external restraining.
6. Curing duration for mass concreting shall be as per ACI 308 Section 3.3.
7. For mass concreting mix design, a mock-up test with following requirements shall be performed to develop a control method statement for the actual pouring at the site:
  - a. Heat dissipation studies in accordance with section 5.4 of ACI 207.1R to confirm that the heat of hydration will not cause a peak temperature and a temperature differential between the interior of the concrete and any outside face greater than the values specified in (5.9, Item 4) and (5.9, Item 5).
  - b. Monitor the heat of hydration (using thermocouples) in trials to demonstrate excessive thermal gradient is not present in the main works.
  - c. Details of cast and test pieces representing the concrete elements (1:1 scale), actual pouring conditions, thickness, formwork and insulation type and removal, ambient radiation and temperature and physical restraints.
  - d. For foundations/ slabs where a 1:1 scale test sample is not possible from construction point of view, a smaller test sample can be tested if the side faces of the samples are provided with 100 mm (4 in) thick polystyrene insulation to simulate the much larger lateral dimension of the actual concrete member, keeping the sample thickness the same as the actual member.
  - e. Monitor strength development versus time (typically at 1, 2, 3, 7 and 28 days).
  - f. Observe bleeding and the general condition of the concrete during the pour.
  - g. Implement protective measurements, insulation and curing to avoid crack formation.
  - h. After removal of formwork, check on crack formation.
  - i. Contractor shall provide to the Principal a detailed report of the results for review and approval.
8. The mix design and control method statements shall be subject to approval from the Principal.
9. When forms are removed before the specified curing is completed, measures shall be taken to continue the curing and provide adequate thermal protection for the concrete.

**5.10 EXTREME WEATHER PROVISIONS****5.10.1 General**

1. Provisions for pouring concrete in extreme weather conditions shall be made in accordance with following Sections (5.10.2) and (5.10.3) as applicable.
2. Provisions for pouring concrete in extreme weather conditions shall be subject to approval by the Principal.

**5.10.2 Hot weather concreting**

1. Concrete placement shall be in accordance with the requirements of ACI 301 and ACI 305R when the ambient temperature exceeds 32 °C (90 °F).

**5.10.3 Cold weather concreting**

1. Concrete placement shall be in accordance with the requirements of ACI 301 and ACI 306R when the ambient temperature drops below 4 °C (40 °F).

**5.11 CURING OF CONCRETE**

1. Concrete curing shall be as per ACI 308R.
2. Unless otherwise specified by the Principal, curing shall continue for a minimum number of cumulative days as specified in Table 3.1 of ACI 308R.
3. The following additional measures shall be provided for curing under cold weather condition:
  - a. Area warming and humidification control of the curing of concrete continues until it develops more than 85 % of the compressive strength.
  - b. Concrete shall be allowed to dry for at least 12 hours prior to discontinuing or removing the temperature protection.
  - c. If freezing or thawing is anticipated at any time during construction, or in service when either the cement paste or the aggregates are critically saturated near the surface of the concrete; use air-entrained concrete in accordance with ACI 318 or ACI 301 or ACI 201.2R.



## 5.12 FINISHING

1. The finish shall be without surface fins, holes or rough patches and as Table 5.3 unless otherwise specified.

**Table 5.3 Concrete surface finish**

| Surface                                | Finish         |
|----------------------------------------|----------------|
| Surface requiring plastering or tiling | Scratched      |
| Area Paving                            | Broom or Belt  |
| Compressor house floors                | Broom or Belt  |
| Cooling tower Basin, Flume             | Floated        |
| Curbs                                  | Broom or Belt  |
| Foundations and pedestal to be grouted | Scratched      |
| Equipment bases                        | Floated        |
| Platforms, Stair Pads, Concrete Stairs | Broom or Belt  |
| Floor Slabs                            | Trowel         |
| Formed Surfaces                        | As-Cast Finish |
| Suction Pits                           | Floated        |

## 5.13 LIQUID RETAINING STRUCTURES

1. Any structure designated as a hydraulic (liquid containing) structure in the drawings shall comply with the following:
  - a. hydraulic structures shall be cured by a wet-cure procedure only (e.g., use of ponding, sprinkling, or a moisture-retaining fabric);
  - b. moist wood forms in contact with concrete shall not be considered as curing for hydraulic structures;
  - c. immediately after placement, concrete shall be protected from premature drying, excessively high or low temperatures (5.10), rain, flowing water or mechanical damage;
  - d. curing shall occur for a minimum of 14 days;
    - i. wall forms shall then be loosened, and water continually sprinkled between the wall and forms.
  - e. the surface of concrete at all joints shall be thoroughly cleaned and all laitance removed before placing adjoining concrete;
  - f. chemically resistant coatings as approved by the Principal shall be used to protect the concrete surface from degradation by harmful liquids;
  - g. liquid retaining structures shall pass a water-tightness test, according to a method statement approved by the Principal.

#### 5.14 PRECAST CONCRETE

1. Connections for precast members shall be in accordance with ACI 550.2R.
2. Precast members shall not be lifted or handled within 72 hours of the completion of pouring and finishing.
3. Structures and members shall be handled, lifted and supported carefully at positions in a manner that does not compromise the integrity of the member.
4. A lifting and erection procedure for precast wall members shall be submitted to the Principal for review.
  - a. Once accepted, no other method shall be used during the whole period of construction, unless specifically approved by the Principal.
5. Connection system between precast structural members shall be approved by the Principal.

#### 5.15 PRE-STRESSED CONCRETE

1. Tendons shall be only procured from plants certified by Post Tensioning Institute (PTI).
2. The tendons and anchorage assemblies shall be cement grouted after post-tensioning per PTI M55.1.

#### 5.16 REPAIR

1. Prior to commencing any concrete work, a repair method statement shall be submitted to the Principal for review and approval.
  - a. The repair materials shall be based on cement types.
  - b. The cured mortar colour shall be approximate to the colour of cured mother material.
2. Concrete surfaces shall not be repaired or touched up after the removal of formwork until the surfaces have been inspected and released in accordance with the quality procedures.

## 6. QUALITY PLANS

### 6.1 GENERAL

1. Written quality plans and procedures shall be submitted to the Principal for review and approval.

*The quality plans and procedures provide details of how compliance with the requirements of this GFS, drawings and Project Specifications is to be achieved.*

2. The quality system shall be based on ISO 9000 quality system requirements.
3. Quality plans shall address all aspects related to local conditions, such as climatic conditions, back up facilities, spare parts, transport possibilities, storage facilities, quarries, local Manufacturer/Suppliers, test facilities (field laboratory).

### 6.2 INSPECTION, TESTING AND REPORTING

1. Inspection and Test Plans (ITPs) identifying the tests, frequencies, rejection criteria and responsibilities shall be prepared as part of the quality plan.
  - a. Tests for approval and acceptance of materials used for concrete production as well as tests on reinforcement shall be performed by a certified testing laboratory approved by the Principal.
2. A reporting system of Quality Control Records (QCRs) shall be part of the quality plan.
3. The reporting system shall record all results obtained during the following:
  - a. inspection of concrete members;
  - b. material and concrete testing;
  - c. material and concrete transporting;
  - d. concrete production;
  - e. concrete supplying;
  - f. concrete placing/pouring;
  - g. concrete finishing;
  - h. concrete curing;
  - i. formwork striking;
  - j. repair work.
4. Weather conditions shall be recorded in the quality records.
5. The proposed ITPs and QCRs shall be included in the Quality Plan and cover the following:
  - a. setting out;
  - b. anchor bolts;
  - c. rebar and cover;
  - d. concrete strength;
  - e. pre-stressing strands;
  - f. inserts and other embedments;
  - g. formwork (including loading capacity, if required);
  - h. concrete placing;
  - i. field concrete workability;

- j. curing;
  - k. removal of formwork (take loading capacity into account);
  - l. joint preparations- defects.
6. Standard forms of inspection and test plans shall be used as a basis for the development of the ITPs and QCRs supplemented by the requirements of the Project Specifications for the work to be executed.
7. After completion, if there is any doubt about the concrete member integrity based on the inspection records, rebound or ultrasonic measurements shall be taken to control and verify the concrete quality against the compliance tests and required concrete quality.
- a. Any deviation shall be reported, including the measurements to be taken.
8. The Principal shall be permitted to make inspections of the following at any time:
- a. at the source of supply of materials;
  - b. at the place of preparation of materials;
  - c. at the mixing plant if ready mixed concrete is used;
  - d. during execution of all concrete work.
9. The inspection and testing agency shall record all test results in a spreadsheet.
- a. The agency shall provide both statistical assessment and graphical trends of the results by each material type.
  - b. This in its native form shall be passed on to the Principal.
  - c. The Contractor shall include in both his weekly and monthly reports.

**7. REFERENCES**

In this GFS, reference is made to the following publications:

NOTES: 1. Unless specifically designated by date, the latest edition of each publication shall be used, together with any amendments/supplements/revisions thereto.

2. The DEPs, SPEPs, GFSs and most referenced external standards are available to Shell staff on the SWW (Shell Wide Web) at <http://sww.shell.com/standards/>.

**SHELL STANDARDS**

|                                                                        |                      |
|------------------------------------------------------------------------|----------------------|
| Procedure for packaged equipment solutions publications                | PES 00.00.00.31-Gen. |
| Packaged equipment solutions feedback form                             | PES 00.00.05.82-Gen. |
| Loading Specification for onshore LNG Facilities                       | GFS IG.34.85.02-Gen. |
| Design and Construction of Steel Structures for Onshore LNG Facilities | GFS IG.34.85.21-Gen. |
| Foundations, Anchor Bolts and Grouting for Onshore LNG Facilities      | GFS IG.34.85.20-Gen. |

**AMERICAN STANDARDS**

|                                                                                    |              |
|------------------------------------------------------------------------------------|--------------|
| Standard Specification for Highway Bridges                                         | AASHTO-7     |
| Specification for Tolerances for Concrete Construction and Materials               | ACI 117      |
| Guide to Durable Concrete                                                          | ACI 201.2R   |
| Guide to Mass Concrete                                                             | ACI 207.1R   |
| Report on Thermal and Volume Change Effects on Cracking of Mass Concrete           | ACI 207.2R   |
| Cooling and Insulating Systems for Mass Concrete                                   | ACI 207.4R   |
| Control of Cracking in Concrete Structures                                         | ACI 224R     |
| Self-Consolidating Concrete                                                        | ACI 237R     |
| Specifications for Structural Concrete                                             | ACI 301      |
| Guide to Concrete Floor and Slab Construction                                      | ACI 302.1R   |
| Placing Concrete by Pumping Methods                                                | ACI 304.2R   |
| Guide for Measuring, Mixing, Transporting, and Placing Concrete                    | ACI 304R     |
| Guide to Hot Weather Concreting                                                    | ACI 305R     |
| Guide to Cold Weather Concreting                                                   | ACI 306R     |
| Guide to External Curing of Concrete                                               | ACI 308      |
| Building Code Requirements for Structural Concrete and Commentary                  | ACI 318      |
| Guide to Formwork for Concrete                                                     | ACI 347R     |
| Code Requirements for Environmental Engineering Concrete Structures and Commentary | ACI 350/350R |
| Report on Fibre Reinforced Concrete                                                | ACI 544.1R-4 |
| Design Guide for Connections in Precast Jointed Systems                            | ACI 550.2R   |
| ACI Detailing Manual                                                               | ACI SP-66    |

|                                                                                                                                                           |                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Standard Specification for Carbon Structural Steel                                                                                                        | ASTM A36/A36M   |
| Standard Specification for Low-Relaxation, Seven-Wire Steel Strand for Prestressed Concrete                                                               | ASTM A416M      |
| Standard Specification for Stress-Relieved Steel Wire for Prestressed Concrete                                                                            | ASTM A421M      |
| Standard Specification for Blended Hydraulic Cement                                                                                                       | ASTM C595/C595M |
| Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement                                                                | ASTM A615/A615M |
| Standard Specification for High-Strength Steel Bars for Prestressed Concrete                                                                              | ASTM A722M      |
| Standard Specification for Steel Strand, Seven-Wire, Uncoated, Compacted for Prestressed Concrete                                                         | ASTM A779M      |
| Standard Specification for Filled Epoxy-Coated Seven-Wire Prestressing Steel Strand                                                                       | ASTM A882M      |
| Standard Specification for Rail-Steel and Axle-Steel Deformed Bars for Concrete Reinforcement                                                             | ASTM A996/A996M |
| Standard Specification for Concrete Aggregates                                                                                                            | ASTM C33        |
| Standard Specification for Ready-Mixed Concrete                                                                                                           | ASTM C94/C94M   |
| Standard Specification for Portland Cement                                                                                                                | ASTM C150/C150M |
| Standard Specification for Chemical Admixtures for Concrete                                                                                               | ASTM C494/C494M |
| Standard Specification for Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use in Concrete                                                          | ASTM C618       |
| Standard Specification for Epoxy-Resin-Base Bonding Systems for Concrete                                                                                  | ASTM C881/C881M |
| Standard Specification for Slag Cement for Use in Concrete and Mortars                                                                                    | ASTM C989       |
| Standard Test Method for Length Change of Hydraulic-Cement Mortars Exposed to a Sulfate Solution                                                          | ASTM C1012      |
| Standard Specification for Preformed Expansion Joint Filler for Concrete (Bituminous Type)                                                                | ASTM D994       |
| Standard Specification for Preformed Expansion Joint Filler for Concrete Paving and Structural Construction (Nonextruding and Resilient Bituminous Types) | ASTM D1751      |
| Standard Specification for Joint and Crack Sealants, Hot Applied, for Concrete and Asphalt Pavements                                                      | ASTM D6690      |
| Standard Test Methods for Water Vapor Transmission of Materials                                                                                           | ASTM E96/E96M   |
| Specification for Grouting of Post-Tensioned Structures                                                                                                   | PTI M55.1       |
| Precast/ Prestressed Concrete Institute - Standard Design Practice                                                                                        | PCI TR-7        |

#### INTERNATIONAL STANDARDS

|                                                          |          |
|----------------------------------------------------------|----------|
| Eurocode 2: Design of concrete structures                | EN 1992  |
| Quality management systems – Fundamentals and vocabulary | ISO 9000 |